DIPLOMA PROGRAMME PRACTICE

Mathematics

Applications and interpretation SL

Set B / Paper 2 / Extended response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

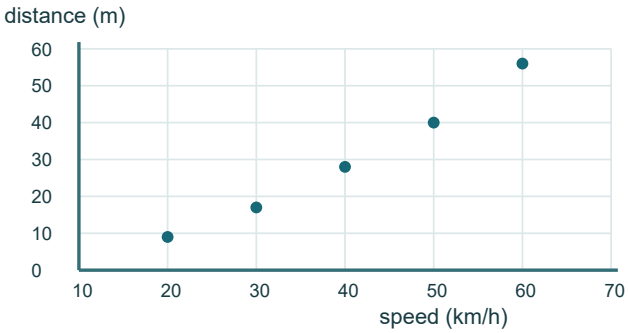
Write your reasoning and working in a separate answer booklet. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

QUESTION 1

20 marks

Five controlled tests on one dry track record speed v (km/h) and stopping distance D (m): (20, 9), (30, 17), (40, 28), (50, 40), (60, 56). The engineer is deciding between a linear regression model L and a second model $Q(v) = 0.20v + 0.012v^2$. Do not round intermediate model coefficients.



(a) Use technology to find the regression line of D on v and Pearson's correlation coefficient. Interpret the direction and strength of association. **[4]**

(b) Calculate observed-minus-predicted residuals for L at speeds 20, 40 and 60. Use their signs to explain a possible weakness of L. **[4]**

(c) A separate test at 70 km/h records 74 m. Compare the absolute percentage prediction errors of L and Q, relative to the measured value. Which model performs better on this test? **[4]**

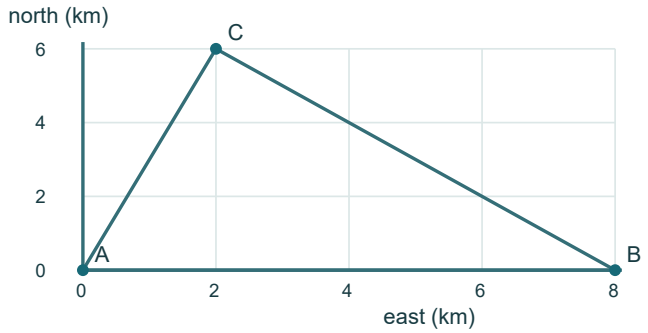
(d) Using Q, calculate the greatest speed at which stopping distance does not exceed 43 m. Give both the mathematical boundary and the greatest safe whole number of km/h. **[4]**

(e) In rain, the quadratic coefficient increases from 0.012 to 0.018 while the linear coefficient remains unchanged. Find the predicted increase in distance at 50 km/h. Explain which coefficient should change instead if only the driver's reaction time increases. **[4]**

QUESTION 2

20 marks

Three emergency beacons have coordinates A(0, 0), B(8, 0) and C(2, 6), in kilometres. The patrol area is the closed triangular region ABC. A distress signal is assigned to its nearest beacon by straight-line distance. Ties may be assigned to either beacon.



(a) Find the perpendicular bisectors of AB and AC. **[4]**

(b) Find the point O equidistant from all three beacons. Calculate that distance and check that O is inside the patrol triangle. **[4]**

(c) Every beacon has a guaranteed range of 4.4 km. A manager asserts there are no uncovered points in the triangle because adjacent coverage disks overlap. Assess the claim using a specific point; explain why overlap is insufficient. **[4]**

(d) A fourth beacon must be at D(k, 1), with $0 \le k \le 8$, and must be within 1.25 km of O. Determine the allowable interval for k. **[4]**

(e) The integer-coordinate site closest to O is chosen. For the point P(5, 3), decide the original nearest beacon or beacons and the new nearest beacon. Could Euclidean assignment ever disagree with the quickest response? Give a reason. **[4]**

QUESTION 3

20 marks

Pouches from a single filling line have independent normally distributed masses with mean 204 g and standard deviation 2.8 g. A pouch is classified as short if it is below 200 g. A fixed batch contains 16 pouches.

(a) Find the probability p that a pouch is short, and the expected number of short pouches in a batch of 16. **[4]**

(b) The batch is held if at least two pouches are short. Calculate the probability that a batch is held. **[4]**

(c) The manager instead specifies that no more than 2% of individual pouches should be short, keeping the same standard deviation. Find the minimum population mean satisfying this target. **[4]**

(d) If the actual short rate is exactly 0.02, determine the largest whole batch size n for which the probability of at least one short pouch is at most 0.30. **[4]**

(e) A blocked nozzle can cause several consecutive shortages. Explain separately how clustering affects the single-pouch model, the expected count if the marginal short rate remains 0.02, and the binomial batch-risk calculation. **[4]**

QUESTION 4

20 marks

A community battery discharges power $P(t) = 2 + 1.2t - 0.2t^2$ kW for a six-hour evening, $0 \le t \le 6$. Energy in kWh is the area under the power graph. Assume that the model repeats on each operating evening unless otherwise stated.

(a) Show that the power has the same value at both endpoints and find its largest value on the interval, stating when it occurs. **[4]**

(b) Find the modelled energy discharged over six hours by integration, and hence the average power. **[4]**

(c) A recorder samples model power only at $t = 0$, 2, 4 and 6. Use the trapezoidal rule on these four samples. Explain the direction of the error without relying only on the numerical difference. **[4]**

(d) The electricity tariff is 0.28 dollars per kWh for the first two hours, then 0.42 dollars per kWh for the next four. Use the continuous energy model to calculate the value of the discharged energy to the nearest cent. **[4]**

(e) A separate seasonal trial delivers 19.2 kWh on night 1 and 0.08 kWh less on each following night. Find total delivery over the first 30 nights and explain why using this model indefinitely would be unreasonable. **[4]**